

Lecture 6 - Potential Outcomes and Average Treatment Effects

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- Potential Outcomes Framework
- Identification of the average treatment effects
 - RCT, A/B testing
 - Selection by Observables
- IV method: constant treatment effect, imperfect compliance -> intent to treat
- (next lecture) Heterogeneous Treatment effects - IV-LATE
 - identify the average treatment effects among compliers (but not always takers/never takers/defiers)

Motivating Example: Head Start Impact Study

- Head Start is a federal early childhood education program for children under the age of 5 from low-income families. The goal of the program is to improve social and cognitive outcomes for low-income preschool children.
- Since its inception in 1965, Head Start has served more than 34 million children. In 2016, it was funded to serve nearly one million children and pregnant women.
- In 1998, the federal government authorized an independent national random assignment study of Head Start to **assess the effects on school readiness and parental outcomes**: The Head Start Impact Study.
 - (Lottery design) 5,000 3- and 4-year olds at oversubscribed programs were randomly assigned to a treatment group (with a Head Start seat) or a control group (without the initial offer of a Head Start seat).
 - the study started in fall 2002 and continued to collect info on students through third grade.

Motivating Example: Head Start Impact Study

- Bitler, Domina, and Hoynes (2015) finds that Head Start leads to large and statistically significant gains in cognitive ability during the pre-school period, but the gains fade out once children enter elementary school. The effects are larger and more persistent for kids at the bottom of the income ladder.



Motivating Example: Head Start Impact Study

Table 1: Summary statistics at baseline

	Baseline	Child Weights	Inv. P-Score Weights
	Control	Difference	Difference
	Mean	T-C	T-C
<i>Child characteristics</i>			
Non-Hispanic white	0.344	-0.022	-0.001
Non-Hispanic black	0.338	0.005	0.001
Hispanic	0.318	0.017	0.0002
Female	0.527	-0.023	0.017
Spanish speaker at home	0.257	0.001	0.016
Low risk	0.789	-0.041*	0.044**
Medium risk	0.156	0.013	-0.019
High risk	0.055	0.028*	-0.025*
Special needs	0.103	0.031**	0.008
Lives with both biological parents	0.499	-0.003	0.024
Urban	0.793	0.001	0.006
<i>Mother/caregiver characteristics</i>			
Non-Hispanic white	0.364	-0.020	-0.008
Non-Hispanic black	0.339	0.001	0.010
Hispanic	0.298	0.018	-0.001
Mother is teenager	0.176	-0.043*	0.023

Potential Outcomes

- Question: what is the causal effect of a treatment on outcomes?
- Let's focus on the classic case with a binary treatment variable: in the data we observe (Y_i, D_i)
- $D_i = 1$ if agent/unit i (in some population of interest) is treated, 0 otherwise.
- Y_i = outcome (r.v.) realized after treatment status is realized.
- Potential outcomes: $(Y_i(0), Y_i(1))$
 - $Y_i(0)$ = outcome for i if not treated ($D_i = 0$)
 - $Y_i(1)$ = outcome for i if treated ($D_i = 1$)
- Treatment effect for unit i :

$$\beta_i := Y_i(1) - Y_i(0) \tag{1}$$

Can you estimate β_i ? Why or why not?

Potential Outcomes

- The fundamental problem is that a unit can either be treated or not, we never observe $(Y_i(1), Y_i(0))$ for the same i simultaneously:

$$Y_i = Y_i(D_i) = D_i Y_i(1) + (1 - D_i) Y_i(0) \quad (2)$$

- *How can we resolve the observability problem?* We need a way to infer Y_{0i} for the units that are treated
- **A possible idea:** use pre-intervention outcomes
 - If manipulation/treatment occurs between *time* $\tau - 1$ and *time* τ , we could assume $y_{i\tau-1}$, the realization of Y for i in period $\tau - 1$ has the same expected value as $Y_i(0)$:

$$E[Y_i(0)] = E[y_{i\tau-1}]$$

Also, post-treatment $y_{i\tau} = Y_i(1)$. Then we can estimate the treatment effect for individual i as:

$$\hat{\beta}_i = y_{i\tau} - y_{i\tau-1}$$

Potential Outcomes

- In lots of cases this works pretty well. E.g., people learn how to drive by seeing how changing the steering wheel changes the direction of the car! BUT there are many cases where “what was happening just before” is not a good counterfactual: those who learn to drive earlier might have more knowledge about cars beforehand!
- This idea will come back to us when we discuss difference-in-difference methods.

Treatment Effects

- Researchers are often interested in the **average treatment effect** in some population of interest:

$$ATE = E[Y_i(1) - Y_i(0)] \quad (3)$$

or the average treatment effect on the treated (focusing on the subpopulation with $D_i = 1$):

$$ATT = E[Y_i(1) - Y_i(0) | D_i = 1] \quad (4)$$

- note ATT is not the same as ATE! There is often selection into the treated subpopulation.
- What we can see in the data are:

$$E[Y_i | D_i = 1] = E[Y_i(1) | D_i = 1]$$

$$E[Y_i | D_i = 0] = E[Y_i(0) | D_i = 0]$$

- What do we get if we compare the outcomes of the treated vs. the outcomes of the untreated (control)?

$$\begin{aligned}\Delta &:= E[Y_i | D_i = 1] - E[Y_i | D_i = 0] \\ &= E[Y_i(1) | D_i = 1] - E[Y_i(0) | D_i = 0] \\ &= \underbrace{E[Y_i(1) - Y_i(0) | D_i = 1]}_{ATT} \\ &\quad + \underbrace{E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0]}_{\text{selection bias}}\end{aligned}\tag{5}$$

- To get Δ closer to ATT (or even ATE), the key is to reduce/eliminate selection bias. How?

Randomized Control Trials

- The most intuitive way to eliminate selection bias is to randomly assign treatment. This is called Randomized Control Trials (RCT).
 - Most papers in development economics rely on RCTs.
 - It's also known as A/B testing in industry. Make sure you mention you've learned this in class in your resume/interviews.
- Draw a random sample from Pop . Within that sample, assign $D_i = 1$ to a **randomly selected** “treatment group” T , and $D_i = 0$ to the remaining “control group” C .
- Define sample means: $\bar{Y}_T = \frac{1}{N_T} \sum_{i \in T} y_i$ among the treated, and $\bar{Y}_C = \frac{1}{N_C} \sum_{i \in C} y_i$ among the control:

$$E[\bar{Y}_T] = E[Y_i(1)|D_i = 1] = E[Y_i(1)] \quad (6)$$

$$E[\bar{Y}_C] = E[Y_i(0)|D_i = 0] = E[Y_i(0)]$$

Why? D_i is randomly assigned:

$$\begin{aligned} & (Y_i(0), Y_i(1)) \perp D_i \\ \Rightarrow & E[Y_i(d)|D_i] = E[Y_i(d)], \text{ for } d \in \{0, 1\} \end{aligned}$$

Thus:

$$E[\overline{Y}_T] - E[\overline{Y}_C] = ATE!!$$

For RCT: we estimate $ATE (=ATT)$, and $SB = 0$.

- Note #1: in RCT we get two things:
 - ① realizations from the **distribution** of $Y_i(1)$ for Pop
 - ② realizations from the **distribution** of $Y_i(0)$ for Pop
- We can estimate $E[Y_i(1)]$, also $median[Y_i(1)]$ and other quantiles for $Y_i(1)$, $Y_i(0)$.
- Recall individual treatment effect is defined as: $\beta_i = Y_i(1) - Y_i(0)$. We have shown that in a RCT we can estimate $ATE = E[\beta_i]$, but what about the median (or, say, 90th percentile) of the distribution of β_i 's? In general not.
 - **Joint** distribution of $(Y_i(1), Y_i(0))$ vs. the distribution of $Y_i(1)$ **and** the dist. of $Y_i(0)$
 - a unit whose $Y_i(1)$ at the 90-th percentile does not necessarily have $Y_i(0)$ at the 90-th percentile.

- Note #2: so far we are describing a “perfect” RCT where
 - we are able to recruit a **random sample** from the population of interest;
 - and treatment status is 100% controlled (all T 's have $D_i = 1$; and C 's have $D_i = 0$), in which case we say we have “**full compliance**”.
- In reality, non-random recruiting and incomplete compliance are big problems.
 - external validity: if the sample in a RCT is non-random, what can we say about the population?
 - internal validity: what if the some units in the control group have learned about the treatment and changed their behaviors? SUTVA (stable unit treatment value assumption) is violated.

Estimate ATE in a Perfect RCT

Analyzing a perfect RCT using a regression model.

- Assume a RCT is well-executed: $(Y_i(0), Y_i(1)) \perp D_i$. How do you obtain a consistent estimate of the ATE?
- One may simply estimate a regression of Y_i on D_i :

$$Y_i = \alpha + \beta D_i + \epsilon_i, \quad (7)$$

which satisfies the FOC: $E[D_i \epsilon_i] = E[\epsilon_i] = 0$,

$$\begin{aligned} \Rightarrow \alpha &= E[Y_i | D_i = 0] \\ \beta &= \underbrace{E[Y_i | D_i = 1] - E[Y_i | D_i = 0]}_{\text{ATE in RCT}} \end{aligned}$$

review lecture 2 - regressions with discrete regressors.

- Sample estimate:

$$\begin{aligned} \hat{\alpha} &= \frac{1}{N_C} \sum_{i: d_i=0} y_i = \bar{Y}_C \\ \hat{\beta} &= \bar{Y}_T - \bar{Y}_C \end{aligned}$$

$\hat{\beta} \xrightarrow{p} \beta$, which is the ATE in a perfect RCT (in contrast with Δ in (5)!)) Note $\hat{\beta}$ is also an unbiased estimate for ATE, $E[\hat{\beta}] = E[\bar{Y}_T - \bar{Y}_C] = E[Y_i(1) - Y_i(0)]$.

Selection on Observables

- Without a RCT, the average treatment effect may also be identified if there are good reasons to believe that the selection into treatment is exogenous **conditional** on observable characteristics of unit i , denoted by X_i that does not depend on treatment D_i .
- Assumptions:

$$(1) \text{ Selection on Observables} \quad (Y_i(0), Y_i(1)) \perp D_i \mid X_i$$

$$(2) \text{ Common Support} \quad \Pr(D_i = 1 \mid X_i) \in (0, 1)$$

- Under (1) and (2), we can estimate the conditional average treatment effect (CATE) first conditional on the values of X :

$$\begin{aligned} & E[Y_i \mid X_i, D_i = 1] - E[Y_i \mid X_i, D_i = 0] \\ &= E[Y_i(1) \mid X_i, D_i = 1] - E[Y_i(0) \mid X_i, D_i = 0] \\ &= E[Y_i(1) \mid X_i] - E[Y_i(0) \mid X_i] \\ &= E[Y_i(1) - Y_i(0) \mid X_i] \end{aligned} \tag{8}$$

Note if (2) is violated, for some X , we only observe treated units, and therefore cannot get sample estimates for $E[Y_i(0) \mid X_i, D_i = 0] \dots$ (the same if we only observe controls)

- From CATE to ATE:

$$\begin{aligned} E[Y_i(1) - Y_i(0)] &= \sum_x Pr(X_i = x) \times E[Y_i(1) - Y_i(0)|X_i = x] \\ &= \int_x E[Y_i(1) - Y_i(0)|X_i = x] dF_X(x) \end{aligned}$$

Estimate ATE under Selection on X

- Suppose there is a finite number of values X can take, it is relatively straightforward: conditional on $X_i = x$, use OLS to estimate CATE,

$$\begin{aligned}\hat{\beta}_{1x} &:= \frac{1}{N_{1x}} \sum_{i: X_i=x, D_i=1} Y_i - \frac{1}{N_{0x}} \sum_{i: X_i=x, D_i=0} Y_i \\ &\xrightarrow{p} E[Y_i|X_i = x, D_i = 1] - E[Y_i|X_i = x, D_i = 0] \\ &= E[Y_i(1) - Y_i(0)|X_i = x]\end{aligned}$$

- However, there may be too many covariates, or continuous variables... You could partition X and impose the assumption that $(Y_i(1), Y_i(0)) \perp D_i | \text{partition of } X$. Alternatively, use propensity scores:

$$p(x) := \Pr(D_i = 1|X_i = x) = E[D_i|X_i = x] \quad (9)$$

and in sample the propensity score of unit i being treated is:

$$p_i := p(x_i).$$

$$E[D_i Y_i] = E[Y_i | D_i = 1] \underbrace{E[D_i]}_{= \Pr(D_i = 1)} \quad D_i \in \{0, 1\}$$

$$E[D_i Y_i | X] = E[Y_i | D_i = 1, X] \cdot \Pr(D_i = 1 | X)$$

Propensity score function $P(x_i)$ (which we note as P_i)

$$D_i = \Pr(D_i = 1 | X_i) = E[D_i | X_i = x]$$

$$Y_i(1) Y_i(0) \perp\!\!\!\perp D_i | X \quad \rightarrow \quad E[Y_i | D_i = 1, X = x] \times P(x)$$

$$\begin{aligned} E\left[\frac{D_i Y_i}{P(x_i)}\right] &= E_x \left[E\left[\frac{D_i Y_i}{P(x_i)} \mid X_i = x\right] \right] \\ &= E_x \left[\frac{E[D_i Y_i | X_i = x]}{P(x)} \right] \end{aligned}$$

$$E[Y(D=1, x)] = \frac{E[Y_i(1) | D_i=1, x]}{E[Y_i(1) | x]}$$

For midterm 2

- Never miss class
- Focus on in class walked-through empirical examples
-

Estimate ATE under Selection on X

- A useful expression when D_i is an indicator:

$$\begin{aligned} E[Y_i|D_i = 1] &= \frac{E[D_i Y_i]}{E[D_i]}, \quad E[Y_i|D_i = 0] = \frac{E[(1 - D_i) Y_i]}{1 - E[D_i]} \\ E[Y_i|D_i = 1, X_i = x] &= \frac{E[D_i Y_i|X_i = x]}{E[D_i|X_i = x]} = \frac{E[D_i Y_i|X_i = x]}{p(x)} \\ E[Y_i|D_i = 0, X_i = x] &= \frac{E[(1 - D_i) Y_i|X_i = x]}{1 - p(x)} \end{aligned} \quad (10)$$

this suggests a consistent estimator for CATE: letting

$N_x = N_{0x} + N_{1x}$ be the num. of units with $X_i = x$, and $\hat{p}(x)$ denote a consistent estimate of $p(x)$,

$$\begin{aligned} &\frac{1}{N_x} \sum_{i: X_i = x} \left(\frac{D_i Y_i}{\hat{p}(x)} - \frac{(1 - D_i) Y_i}{1 - \hat{p}(x)} \right) \\ &\xrightarrow{p} \frac{E[D_i Y_i|X_i = x]}{p(x)} - \frac{E[(1 - D_i) Y_i|X_i = x]}{1 - p(x)} \\ &= E[Y_i(1) - Y_i(0)|X_i = x] \sim \text{CATE!!!} \end{aligned} \quad (11)$$

Estimate ATE under Selection on X

- Now let's apply the law of iterated expectations to the following:

$$\begin{aligned}E\left[\frac{D_i Y_i}{p_i}\right] &= E_X[E[\frac{D_i Y_i}{p(x)}|X_i = x]] \\&= E_X[E[Y_i|D_i = 1, X_i = x]] \\&= E_X[E[Y_i(1)|X_i = x]] = E[Y_i(1)]\end{aligned}$$

and similarly,

$$E\left[\frac{(1 - D_i) Y_i}{1 - p_i}\right] = E_X[E[\frac{(1 - D_i) Y_i}{1 - p(x)}|X_i = x]] = E[Y_i(0)]$$

Show:

$$\frac{1}{N} \sum_i \frac{D_i Y_i}{\hat{p}_i} - \frac{(1 - D_i) Y_i}{(1 - \hat{p}_i)} \xrightarrow{p} ATE \quad (12)$$

- This implies instead of estimating CATE for every possible value of X , we can estimate the propensity score $\hat{p}_i = \hat{Pr}(D_i = 1|X_i = x) \xrightarrow{p} E[D_i|X_i = x]$ first, and then create $\frac{D_i Y_i}{\hat{p}_i}, \frac{(1-D_i) Y_i}{(1-\hat{p}_i)}$, the difference in the sample means would be a consistent estimate for ATE.

Propensity Scores (Bonus)

- Rosenbaum and Rubin(1983): The selection on observable assumption implies the following exogenous selection on propensity scores

$$(Y_i(0), Y_i(1)) \perp D_i | X_i \rightarrow (Y_i(0), Y_i(1)) \perp D_i | p(X_i)$$

- To prove this, it is enough to show

$$E[D_i | Y_i(0), Y_i(1), p] = E[D_i | p] = p \quad (13)$$

- ① By Law of Iterated Expectations (over X conditional on $p(X) = p$):

$$\begin{aligned} E[D_i | Y_i(0), Y_i(1), p] &= E_X[E[D_i | Y_i(0), Y_i(1), X, p(X) = p]] \\ &= E_X[E[D_i | Y_i(0), Y_i(1), X, p(X) = p]] \\ &= E_X[\underbrace{E[D_i | Y_i(0), Y_i(1), X]}_{p(X)} | p(X) = p] \end{aligned}$$

- ② $(Y_i(0), Y_i(1)) \perp D_i | X \Rightarrow E[D_i | Y_i(0), Y_i(1), X] = E[D_i | X] = p(X)$

- ③ therefore

$$E[D_i | Y_i(0), Y_i(1), p] = E_X[p(X) | p(X) = p] = p!!$$

IV for ATE: Constant Treatment Effect

- RCT may not have complete **compliance**. For example, consider the Head Start program that provides subsidized pre-school for young kids of poor families. Head Start Impact Study is a RCT: randomization (using lottery). Applicants for HS assigned to program group (offered HS spot) or control group (denied).
 - Outcome Y_i : cognitive test score of kid after 1 year
 - Endogenous variable of interest: $D_i = 1$ if the kid attended preschool at age 3.
 - (Instrument) Random assignment: $Z_i = 1$ if the kid is assigned to the treated group (receive HS subsidy) \
- Note $E[D_i|Z_i = 1] < 1$ and $E[D_i|Z_i = 0] > 0$.

Z_i is whether offered head start group ($Z_i=1$)

Table 2: Child care setting, by treatment or control status

	Treatment Mean	Control Mean	Difference T-C
<u>Head Start in HS Year</u>			
Head Start (Administrative report)	0.857	0.153	0.705***
<u>Parent Report, Spring 2003</u>			
Head Start	0.823	0.146	0.677***
Other Center	0.068	0.252	-0.183***
Family day care	0.014	0.064	-0.050***
Parent/relative	0.094	0.536	-0.442***
Not reported	0.001	0.002	-0.001

would be 1 if perfect compliance

the people not offered but took up treatment

whereas $D_i=1$ is your kid actually taking a seat in the head start program

b/c ethical guidelines we don't have perfect RCT $\Rightarrow Z_i$ and D_i

IV for ATE: Constant Treatment Effect

26:02

- It's still possible to identify ATE if the underlying causal model features a constant treatment effect:

$$\forall i: Y_i(1) = Y_i(0) + \beta_1$$

this is still $Y_i(d)$

if you have $Y_i(D_i, Z_i) = Y_i(0)$
exclusion restriction

- Causal model (w constant TE):

$$Y_i = \underbrace{Y_i(0)}_{E \rightarrow \beta_0} + D_i \times \underbrace{(Y_i(1) - Y_i(0))}_{\beta_1}$$

Z_i : instrument for D_i
($cov(Z_i, D_i) \neq 0$)

* assume no spillover
(i.e. people w/ offer not
selling to someone who
didn't)
b/c

35:00 →

$$= \beta_0 + \beta_1 D_i + \underbrace{u_i}_{\text{b/c } cov(D_i, Y_i(0)) \neq 0 \Rightarrow \text{we add } u_i} = [Y_i(0) - E[Y_i(0)]]$$

in which $\beta_1 \equiv Y_i(1) - Y_i(0)$, $\beta_0 = E[Y_i(0)]$. We want to get a consistent estimate of β_1 , which is the ATE in this special case with constant treatment effect. What about OLS of Y_i on $(1, D_i)$?

40:00

$$\hat{\beta}_1^{OLS} = \frac{1}{N_1} \sum_{i:D_i=1} Y_i - \frac{1}{N_0} \sum_{i:D_i=0} Y_i \quad (14)$$

$$= \frac{1}{N_1} \sum_{i:D_i=1} (Y_i(1) - Y_i(0)) + \left(\frac{1}{N_1} \sum_{i:D_i=1} Y_i(0) - \frac{1}{N_0} \sum_{i:D_i=0} Y_i(0) \right)$$

→ Review w/ OLS
w/ dummies

$$\xrightarrow{P} \underbrace{E[Y_i(1) - Y_i(0) | D_i = 1]}_{=\beta_1} + \underbrace{(E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0])}_{\text{selection bias}}$$

IV for ATE: Constant Treatment Effect

- IV: Head Start subsidy is randomly assigned, $Z_i \perp (Y_i(0), Y_i(1))$ is an instrument for actually attending pre-school D_i :

$$\text{First stage: } D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (15)$$

$$\pi_0 = E[D_i | Z_i = 0]$$

$$\pi_1 = E[D_i | Z_i = 1] - E[D_i | Z_i = 0]$$

- If Head Start is a RCT with full compliance, $\pi_0 = 0$ and $\pi_1 = 1$. Otherwise, there can be families who receive subsidy but still not sending kids to pre-school ($\pi_1 < 1$), or families who use pre-school without the subsidy ($\pi_0 > 0$).

which can still be correlated with both

control group

treated group

IV for ATE: Constant Treatment Effect

- Second stage: plug the first stage into the causal model:

49:01

$$\begin{aligned} Y_i &= \beta_0 + \beta_1(\pi_0 + \pi_1 Z_i + \eta_i) + u_i \\ &= \beta_0 + \beta_1(\pi_0 + \pi_1 Z_i) + (\beta_1 \eta_i + u_i) \end{aligned}$$

since $E[Z_i \times (\beta_1 \eta_i + u_i)] = 0$ by random assign/exclusion restriction, we can estimate a reduced-form regression of Y_i on $(1, Z_i)$, and obtain a consistent estimate for $\beta_1 \times \pi_1$:

$$\begin{aligned} Y_i &= \hat{\delta}_0 + \hat{\delta}_1 Z_i + \hat{\epsilon}_i \\ \hat{\beta}_1^{IV} &= \hat{\delta}_1 / \hat{\pi}_1 \xrightarrow{p} \beta_1 \end{aligned}$$

where $\beta_1 \equiv Y_i(1) - Y_i(0)$ is the **constant** treatment effect in the population.

IV for ATE: Constant Treatment Effect

- In summary, in the special case with constant treatment effect, $Y_i(1) - Y_i(0) \equiv \beta_1$, we can get a consistent estimate for β_1 with a valid instrument for (endogenous) D_i . So even if the RCT does not have full compliance, we can still estimate the ATE under the assumption of constant treatment effect.
- Question: what does the IV estimate represent if $Y_i(1) - Y_i(0)$ vary across individuals?

Imperfect Compliance and Intent to Treat

- Define $N_1 = \sum_{i=1}^N Z_i$ (num. units assigned to treatment, “lottery winners”), $N_0 = \sum_i (1 - Z_i)$ (num units assigned to control).
 $\bar{D}_1 = \frac{1}{N_1} \sum_{i: Z_i=1} D_i$: the average of D_i among lottery winners, and
 $\bar{Y}_1 = \frac{1}{N_1} \sum_{i: Z_i=1} Y_i$: the average outcome lottery winners.
- Let's take a closer look at the first-stage:

$$D_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i + \eta_i$$

$$\hat{\pi}_1 = \bar{D}_1 - \bar{D}_0. ; \hat{\pi} = \bar{D}_i \text{ if perfect compliance}$$

$\hat{\pi}_1$ represents **compliance**: the effect of HS offer (winning the lottery) on the actual enrollment in Head Start.

- In the Head Start Impact Study, $\hat{\pi}_0 = 0.153$, $\hat{\pi}_1 = 0.705$. (“cross-over and incomplete take-up”)
- Reduced Form:

$$Y_i = \hat{\delta}_0 + \hat{\delta}_1 Z_i + \epsilon_i$$

$$\hat{\delta}_1 = \bar{Y}_1 - \bar{Y}_0.$$

$\hat{\delta}_1$ represents **intent to treat (ITT)**: the impact of HS offer (winning the lottery) on the outcome of kids.

51:40

2SLS: Local Average Treatment Effects

- 53:30 • Second stage (2SLS):

$$\hat{Y}_i = \hat{\beta}_0^{iv} + \hat{\beta}_1^{iv} \hat{D}_i$$
$$\hat{\beta}_1^{iv} = \frac{\hat{\delta}_1}{\hat{\pi}_1} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{D}_1 - \bar{D}_0}.$$

Handwritten annotations: 0.85 under $\bar{D}_1$, 0.15 under $\bar{D}_0$, and a blue arrow pointing from the denominator to the text "LATE" in the following paragraph.

$\hat{\beta}_1^{iv}$ represents the **local average treatment effect (LATE)** $E[Y_i(1) - Y_i(0) | \text{Compliers}]$, the average causal effect of HS **enrollment** among compliers whose treatment (enrollment) is induced by the instrument (lottery).

- Note in program evaluation literature, the IV estimate may be called the treatment-on-the-treated "TOT" $E[Y_i(1) - Y_i(0) | D_i = 1]$.
 - TOT=LATE if units in the control group cannot be treated - "1-sided non-compliance". Everyone with $D = 1$ is a complier.

reduced form in sample:

$$Y_i = \hat{\delta}_0 + \hat{\delta}_1 Z_i + \gamma_i \quad ; \quad Z_i \in \{0, 1\}$$

$$\hat{\delta}_0 = \bar{Y}_0 = \frac{1}{N_0} \sum_{i: Z_i=0} Y_i$$

$$\hat{\delta}_1 = \bar{Y}_1 - \bar{Y}_0 = \left(\frac{1}{N_1} \sum_{i: Z_i=1} Y_i \right) - \bar{Y}_0$$

2SLS: Local Average Treatment Effects

Table 4: Reduced-form estimates of the effect of an offer of a Head Start slot, first stage estimates of the effect of an offer on administrative Head Start take-up, and two-stage least squares estimates (TOT) of the effect of attending Head Start on PPVT, by demographic subgroup for the Head Start year (spring 2003)

	Control mean [SD]	Reduced form (SE)	First stage (SE)	Two-stage least (SE)
<i>Race/ethnicity subgroups</i>				
Hispanic	234	11.54*** (3.06)	0.673*** (0.060)	17.13*** (4.94)
Non-Hispanic Black	250	5.19** (2.58)	0.694*** (0.046)	7.48* (3.83)
Non-Hispanic White	268	6.49** (3.14)	0.749*** (0.033)	8.66** (4.23)
<i>Language subgroups</i>				
Spanish speaker at home	223	15.0*** (3.30)	0.672*** (0.057)	22.37*** (5.65)
English speaker at home	261	4.93*** (1.86)	0.717*** (0.029)	6.87*** (2.62)
<i>Baseline PPVT score tercile subgroups</i>				
Bottom tercile, baseline PPVT	229	11.5*** (2.77)	0.787*** (0.035)	14.2*** (3.72)
Middle tercile, baseline PPVT	251	2.21 (2.45)	0.641*** (0.047)	3.44 (3.84)
Top tercile, baseline PPVT	274	7.50** (3.16)	0.674*** (0.041)	11.13** (4.78)

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